zoro.com
Keep Your Budget Balanced.
Shop Now

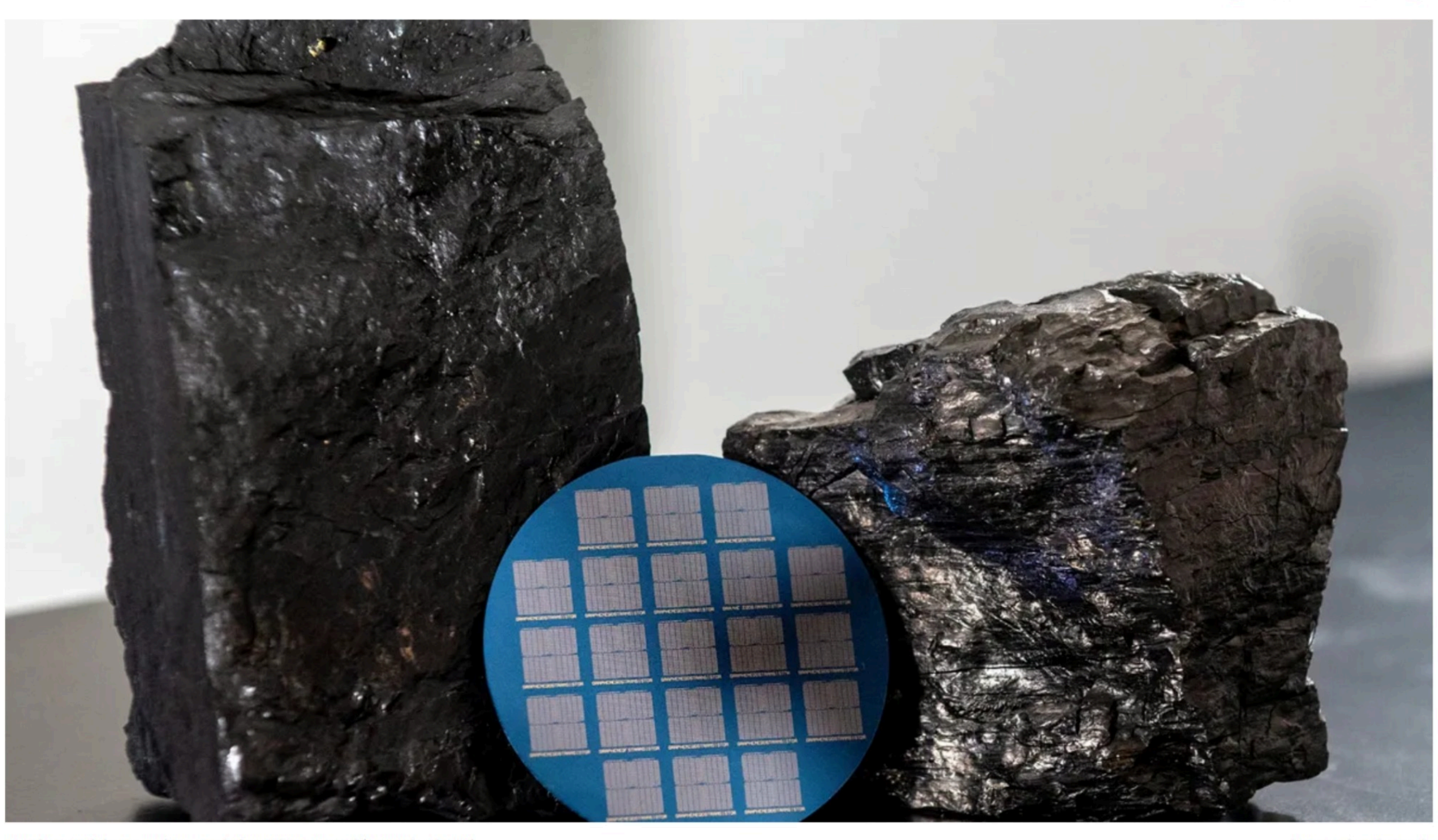
New study discovers coal's unexpected role in next-gen microelectronics

Discover how researchers leverage coal's properties to craft ultra-thin materials, enabling superior performance in next-gen electronic devices.

Abdul-Rahman Oladimeji Bello

Published: Jan 04, 2024 01:30 PM EST

SCIENCE



A wafer containing memristors made from Blue Gem coal (2 samples shown)

better-microelectronic

Get a daily digest of the latest news in tech, science, and technology, delivered right to your mailbox. Subscribe now.

Email address

Join Now

By subscribing, you agree to our Terms of Use and Policies You may unsubscribe at any time.

Coal is synonymous with fossil fuel consumption and environmental impact. It emerges as an unlikely hero in the realm of advanced microelectronics.

Collaborative research efforts led by the University of Illinois Urbana-Champaign, in conjunction with the National Energy Technology Laboratory (NETL), Oak Ridge National Laboratory, and the Taiwan Semiconductor Manufacturing Company, have unlocked the transformative potential of coal.

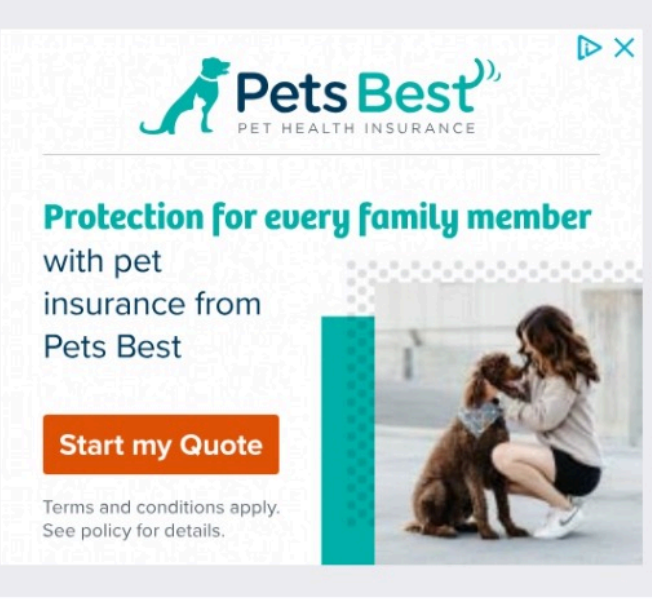
That is, they have demonstrated coal's ability to yield high-purity materials mere atoms thick, offering superior performance for cutting-edge electronic devices.

Professor Qing Cao, a prominent figure in this pioneering initiative, underscores the revolutionary aspect: "Our novel processing techniques have transformed coal—traditionally viewed as bulky and environmentally harmful—into high-purity materials that are incredibly thin.

SEE ALSO

RELATED

- China tests world's largest, 600,000-ton, coal-to-ethanol production plant
- How coal power plants can go green with Carnot batteries
- Coal: A renewable resource? Unlocking hydrogen storage possibilities



Pets Best
PET HEALTH INSURANCE
Protection for every family member with pet insurance from Pets Best
Start my Quote
Terms and conditions apply. See policy for details.

These materials have unlocked possibilities for creating ultra-small electronics with exceptional performance characteristics."

The NETL's groundbreaking process converts coal char into minuscule nanoscale carbon disks termed "carbon dots." These atomically thin carbon dots form the foundation for creating membranes essential in advanced electronic technologies, notably two-dimensional transistors and memristors, crucial components for the next generation of microelectronics.

In the relentless pursuit of smaller, faster, and more energy-efficient electronics, the ultimate goal is to fashion devices that are only one or two atoms thick.

While significant strides have been made in studying ultrathin semiconductors, the necessity for atomically thin insulators—a fundamental requirement in constructing transistors and memristors—has become increasingly apparent.

Save up to 90% on Temu
Temu

Enter the coal-derived carbon layers, exhibiting promising traits as efficient insulators. Professor Cao's research group demonstrated their application in two-dimensional transistors, achieving operating speeds that are over two times faster while reducing energy consumption.

These amorphous coal-derived carbon layers prevent leakage currents, ensuring enhanced device performance and operational efficiency.

Unleashing AI potential: memristors at the forefront

Delving further into memristors—an essential technology for advancing Artificial Intelligence—researchers observed that coal-derived carbon layers facilitated faster filament formation with significantly lower energy consumption.

This development accelerates operations while ensuring improved data storage reliability. It, therefore, enhances AI technology's capabilities and implementation.

While proof-of-concept devices using coal-derived carbon layers showcase immense promise, the real challenge lies in scaling up production for practical application.

Professor Cao emphasizes the collaborative nature of their work with industry leaders like Taiwan Semiconductor, recognizing the growing interest in two-dimensional devices. The primary objective remains the development of industrial-scale processes for coal-based carbon insulators, bridging the gap between groundbreaking research and large-scale industrial implementation.

This collaborative effort represents a paradigm shift in perceiving coal—transforming it from a traditional energy resource to a cutting-edge material driving innovation in microelectronics.

As coal's unexpected role unfolds in shaping the future of electronics, this groundbreaking research paves the way for a sustainable, efficient, and remarkably innovative electronic landscape.

The study's results were published in Nature *(Communications Engineering)* and can be found here.

HOME > SCIENCE

Add Interesting Engineering to your Google News feed. FOLLOW ON Google News

SHOW COMMENT (0)

FOR YOU

UK invests in domestic
Trench 94: The eerie
Male contraceptive
The 100-year-old
How en

JOB BOARD

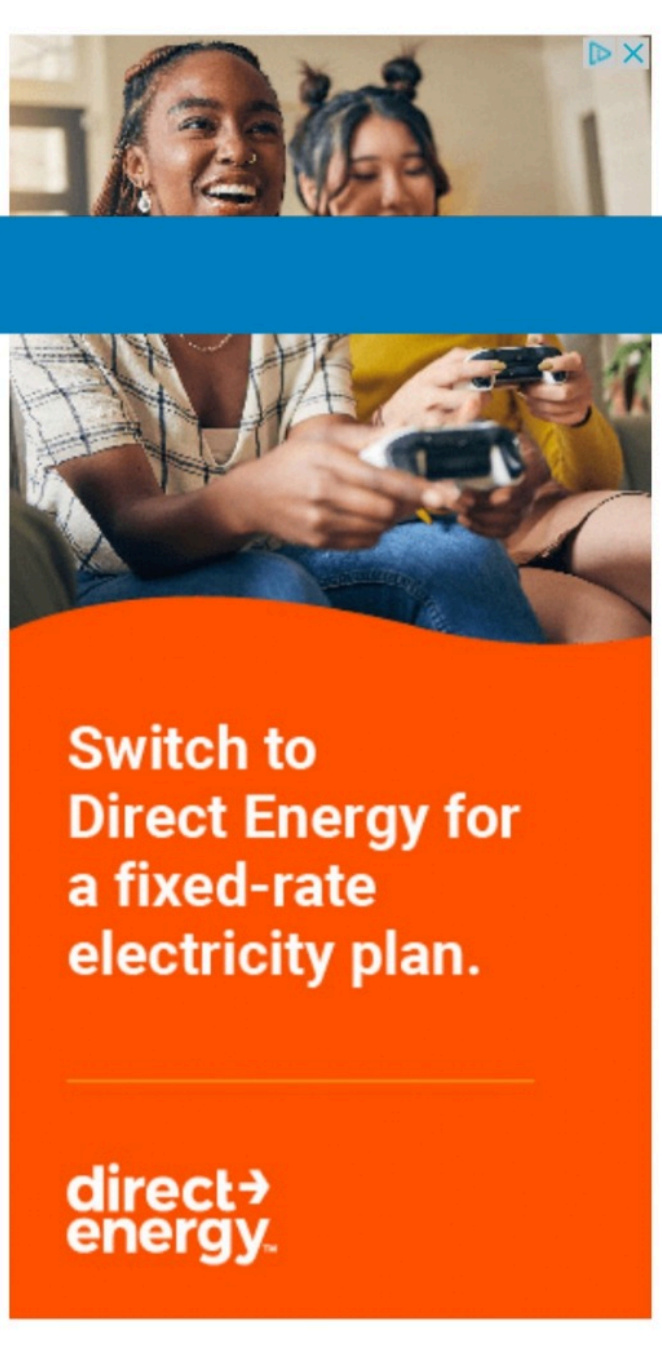
Capgemini
Mechanical & Physical Engineer - Core Engine...
Auburn Hills
See Job

Blue Origin, LLC
Reliability (RAMS) Engineer - Engines...
Huntsville
See Job

Honeywell
Senior Advanced Project Engineer-Turbofan...
Phoenix
See Job

Arcadis
Environmental Engineer/ Civil Engineer
Syracuse
$52,000 - $78,000 ...
See Job

Search More Roles



Switch to Direct Energy for a fixed-rate electricity plan.
direct energy

SPONSORED CONTENT

Recommended by Outbrain

Illinois Find Out How Others in Champaign Are Getting Weed Delivered
HelloMood

ED is Not From Low Testosterone: Meet the Re...
alphahealthfindings.com

Scientist Reveals Why Ringing In The Ears or...
Health Truth Finder

A Game Where Everything is Possible
BuzzDaily Winners

Star Trek Fleet Command Is Now on Desktop. Downloa...
Star Trek Fleet Command

The New York Times
~~$6.25~~ $1 a week for your first six months.
SUBSCRIBE NOW
Cancel or pause anytime.

If You Have Toenail Fungus Try This Tonight (It's Genius!)
wellnessguide101.com

This game look so real that it's worth playing just to see it!
The only official free GoT game! No installation is required!
Game of Thrones

This Drone Shot Cost The Photographer Dearly
Investing Magazine

Forget Furosemide, Use This Household Item To Help Drain...
healthlabnews.com

1/2 Cup Of This (Before Bed) Eats Your Belly Fat Like Never Before!
getfittoday.online

[Gallery] They Stole The Show At Burning Man Festival
DailyChoices

Disney Bundle
Access epic stories, tons of movies & shows, and more — all for one...
Disney+

[Pics] Here's What 30 Historical Figures Actually Looked Like In...
Investing Magazine

Brain Surgeon: "Tinnitus is Not From Low Vitamin B. Meet the...
The Close Relationship Between Stress and Tinnitus
healthtrend.live

Game shows what the world

The #1 Toilet Cleaning Hack I

Pluto - Indica Hybrid - 24.98%